

Calculus and analysis — Session 1

Vahid Shahverdi

TU/e IMC team

Rolle's theorem and the mean value theorem

Throughout, let $f \in C([a, b])$ be differentiable on (a, b) , where $a < b$.

Rolle's theorem

If $f(a) = f(b)$, then there exists $c \in (a, b)$ with

$$f'(c) = 0.$$

Rolle's theorem and the mean value theorem

Throughout, let $f \in C([a, b])$ be differentiable on (a, b) , where $a < b$.

Rolle's theorem

If $f(a) = f(b)$, then there exists $c \in (a, b)$ with

$$f'(c) = 0.$$

Lagrange's mean value theorem

There exists $c \in (a, b)$ with

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Apply Rolle to $f(x) - f(a) - \frac{f(b) - f(a)}{b - a}(x - a)$.

Cauchy's mean value theorem

Cauchy's mean value theorem

Let $f, g \in C([a, b])$ be differentiable on (a, b) . There exists $c \in (a, b)$ such that

$$(g(b) - g(a))f'(c) = (f(b) - f(a))g'(c).$$

Cauchy's mean value theorem

Cauchy's mean value theorem

Let $f, g \in C([a, b])$ be differentiable on (a, b) . There exists $c \in (a, b)$ such that

$$(g(b) - g(a))f'(c) = (f(b) - f(a))g'(c).$$

Apply Rolle's theorem to

$$H(x) = (g(b) - g(a))f(x) - (f(b) - f(a))g(x).$$

We have $H(a) = H(b)$.

This form requires no division by $g'(c)$.

Auxiliary functions

Target expression	Derivative to use
$f' - \lambda f$	$(e^{-\lambda x} f)' = e^{-\lambda x} (f' - \lambda f)$
$xf' - f$	$(f/x)' = \frac{xf' - f}{x^2}$
$2f' + xf''$	$(xf)'' = 2f' + xf''$
$f'' + 2xf' + (x^2 + 1)f$	$(e^{x^2/2} f)'' = e^{x^2/2} (f'' + 2xf' + (x^2 + 1)f)$

The auxiliary function must also fit the endpoint conditions.

Example 1: IMC 2025, Problem 6

Let $f \in C^1((0, \infty))$, and let $0 < a < b$ satisfy

$$f(a) = f(b) = k.$$

Prove that there exists $c \in (a, b)$ such that

$$f(c) - cf'(c) = k.$$

Example 1: IMC 2025, Problem 6

Let $f \in C^1((0, \infty))$, and let $0 < a < b$ satisfy

$$f(a) = f(b) = k.$$

Prove that there exists $c \in (a, b)$ such that

$$f(c) - cf'(c) = k.$$

Auxiliary function

$$H(x) = \frac{f(x) - k}{x}.$$

Example 1: solution

For

$$H(x) = \frac{f(x) - k}{x},$$

we have $H(a) = H(b) = 0$.

By Rolle's theorem, $H'(c) = 0$ for some $c \in (a, b)$.

Example 1: solution

For

$$H(x) = \frac{f(x) - k}{x},$$

we have $H(a) = H(b) = 0$.

By Rolle's theorem, $H'(c) = 0$ for some $c \in (a, b)$.

$$0 = H'(c) = \frac{cf'(c) - f(c) + k}{c^2}.$$

Hence

$$f(c) - cf'(c) = k.$$

Exercise: IMC 2023, Problem 7

Find all $\alpha \in \mathbb{R}$ such that every continuous function

$$f : [0, 1] \longrightarrow \mathbb{R}, \quad f(0) = 0, \quad f(1) = 1,$$

differentiable on $(0, 1)$, satisfies

$$f'(c) = f(c) + \alpha$$

for some $c \in (0, 1)$.

Exercise: IMC 2023, Problem 7

Find all $\alpha \in \mathbb{R}$ such that every continuous function

$$f : [0, 1] \longrightarrow \mathbb{R}, \quad f(0) = 0, \quad f(1) = 1,$$

differentiable on $(0, 1)$, satisfies

$$f'(c) = f(c) + \alpha$$

for some $c \in (0, 1)$.

Hint: first test a function for which $f' - f$ is constant.

Notes: I2023-2-7. IMC 2023, Day 2, Problem 7.

IMC 2023: determine the constant

Consider

$$h(x) = \frac{e^x - 1}{e - 1}.$$

Then

$$h(0) = 0, \quad h(1) = 1, \quad h'(x) - h(x) = \frac{1}{e - 1}.$$

IMC 2023: determine the constant

Consider

$$h(x) = \frac{e^x - 1}{e - 1}.$$

Then

$$h(0) = 0, \quad h(1) = 1, \quad h'(x) - h(x) = \frac{1}{e - 1}.$$

Any constant that works for every f must therefore be

$$\alpha = \frac{1}{e - 1}.$$

IMC 2023: prove existence

For an arbitrary f with the prescribed endpoints, set

$$H(x) = e^{-x}(f(x) - h(x)), \quad h(x) = \frac{e^x - 1}{e - 1}.$$

We have $H(0) = H(1) = 0$.

IMC 2023: prove existence

For an arbitrary f with the prescribed endpoints, set

$$H(x) = e^{-x}(f(x) - h(x)), \quad h(x) = \frac{e^x - 1}{e - 1}.$$

We have $H(0) = H(1) = 0$.

Rolle's theorem gives

$$0 = H'(c) = e^{-c} \left(f'(c) - f(c) - \frac{1}{e - 1} \right).$$

Thus the answer is

$$\left\{ \frac{1}{e - 1} \right\}.$$

Weighted Rolle's lemma

Weighted Rolle

Suppose f is differentiable on an interval and has $m \geq 2$ distinct zeros. For every $\lambda \in \mathbb{R}$,

$$f' - \lambda f$$

has at least $m - 1$ distinct zeros between them.

Weighted Rolle's lemma

Weighted Rolle

Suppose f is differentiable on an interval and has $m \geq 2$ distinct zeros. For every $\lambda \in \mathbb{R}$,

$$f' - \lambda f$$

has at least $m - 1$ distinct zeros between them.

Proof. Between consecutive zeros, apply Rolle to

$$H(x) = e^{-\lambda x} f(x), \quad H'(x) = e^{-\lambda x} (f'(x) - \lambda f(x)).$$

The intervals between consecutive zeros are disjoint.

Darboux's theorem

Darboux's theorem for derivatives

If H is differentiable on an interval, then H' has the intermediate value property.

In particular, if

$$H'(a) < \lambda < H'(b),$$

then $H'(c) = \lambda$ for some $c \in (a, b)$.

Darboux's theorem

Darboux's theorem for derivatives

If H is differentiable on an interval, then H' has the intermediate value property.

In particular, if

$$H'(a) < \lambda < H'(b),$$

then $H'(c) = \lambda$ for some $c \in (a, b)$.

Proof. Set $K(x) = H(x) - \lambda x$. Since

$$K'(a) < 0 < K'(b),$$

neither endpoint minimizes K on $[a, b]$. At an interior minimum, $K'(c) = 0$.

Continuity of H' is not required.

Example 2: IMC 2019, Problem 6

Let $f, g : \mathbb{R} \rightarrow \mathbb{R}$ be continuous, with g differentiable. Suppose

$$(f(0) - g'(0))(g'(1) - f(1)) > 0.$$

Prove that there exists $c \in (0, 1)$ such that

$$f(c) = g'(c).$$

Notes: I2019-2-6. IMC 2019, Day 2, Problem 6.

Example 2: solution

Define

$$H(x) = \int_0^x f(t) dt - g(x).$$

By the fundamental theorem of calculus,

$$H'(x) = f(x) - g'(x).$$

Example 2: solution

Define

$$H(x) = \int_0^x f(t) dt - g(x).$$

By the fundamental theorem of calculus,

$$H'(x) = f(x) - g'(x).$$

The hypothesis gives

$$H'(0)H'(1) < 0.$$

Darboux's theorem therefore yields

$$H'(c) = 0, \quad \text{so} \quad f(c) = g'(c).$$

The ordinary intermediate value theorem would require continuity of $f - g'$, which is not assumed.

Homework: Putnam 2015, B1

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be three-times differentiable and have at least five distinct real zeros.

Prove that

$$f + 6f' + 12f'' + 8f^{(3)}$$

has at least two distinct real zeros.

Homework: Putnam 2015, B1

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be three-times differentiable and have at least five distinct real zeros.

Prove that

$$f + 6f' + 12f'' + 8f^{(3)}$$

has at least two distinct real zeros.

Hint: factor the differential operator.

Notes: P2015-B1. Putnam 2015, B1.

Putnam 2015: solution

Write $D = \frac{d}{dx}$. Then

$$(1 + 2D)^3 f = f + 6f' + 12f'' + 8f^{(3)}.$$

Putnam 2015: solution

Write $D = \frac{d}{dx}$. Then

$$(1 + 2D)^3 f = f + 6f' + 12f'' + 8f^{(3)}.$$

Since

$$(1 + 2D)u = 2 \left(u' + \frac{1}{2}u \right),$$

weighted Rolle's lemma applies with $\lambda = -\frac{1}{2}$.

Putnam 2015: solution

Write $D = \frac{d}{dx}$. Then

$$(1 + 2D)^3 f = f + 6f' + 12f'' + 8f^{(3)}.$$

Since

$$(1 + 2D)u = 2 \left(u' + \frac{1}{2}u \right),$$

weighted Rolle's lemma applies with $\lambda = -\frac{1}{2}$.

Function	Number of distinct zeros
f	≥ 5
$(1 + 2D)f$	≥ 4
$(1 + 2D)^2 f$	≥ 3
$(1 + 2D)^3 f$	≥ 2